



Clonucopya

Clonal Evolution & Treatment Report

Study Statistics: dream_P10

Overview of study dream_P10's key statistics by sample (N = 1). This section summarizes the core characteristics of each sample in the study and serves as the starting point for interpreting the rest of the report. The values reflect each sample's mutational burden, genomic alterations, and the number of therapeutic options identified (total drugs and Best Therapeutic Candidates, BTC). The exact set of metrics shown may vary slightly depending on the analysis settings selected for this run. Refer to the table below for the specific indicators reported.

This report was generated in clinical mode. In clinical mode, results are filtered more stringently, excluding drugs with experimental status and pathway member interaction type. In discovery mode, no filters are applied, and all drug hits identified are displayed regardless of their experimental status or interaction type.

All definitions are provided in the Glossary of Terms.

The source files are available at: results/dream_P10/report/components/report_panel*.tsv.

Sample Name tumour	Sex unknown	Tumor Content 0.95	Total Drugs 143
SNVs/Indels 8192	CNAs 75	SNV-CNA Intersects 8069	BTCs 0

Tumor Phylogeny

Based on the variant allele frequency (VAF) and changes in copy number, the evolutionary hierarchy of the tumor has been inferred. The phylogenetic tree illustrates the relationships among the different cellular subpopulations (clones).

The source files are available at: results/dream_P10/report/components/clonal_tree.png.

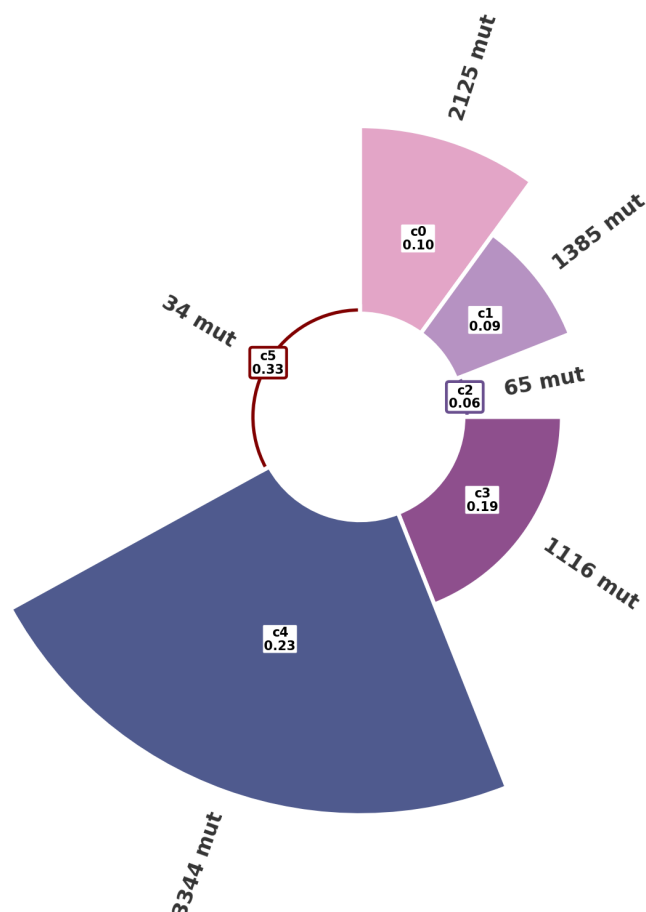


Clonal composition

This is a rose (nightingale) chart showing the clonal and mutational composition at the study level. Each bar represents one clone: its width reflects clonal prevalence (the clone's weight within the tumor's overall clonal structure), and its height reflects the number of mutations that define that clone's identity. Note that prevalence and mutation count are not directly correlated. A clone can carry many mutations yet represent only a small fraction of the tumor, or vice versa, depending on each mutation's biological impact.

The prevalence values across clones may not sum to 100%. This gap represents the undetermined portion of the clonal structure, which can result from normal-cell contamination (already accounted for via the tumor purity estimate), statistical uncertainty in prevalence estimation, or rounding.

The source files are available at: results/dream_P10/report/components/study_name}_clonal_composition.png.



Drug Summary

The Drug Summary provides a high-level overview of the therapeutic candidates identified across the entire study. For each drug, the table reports the number of genetic alterations supporting its prioritization, its regulatory approval status (APPROVED, CLINICAL_TRIALS, or EXPERIMENTAL), and the type of interaction with the affected genes (DIRECT TARGET, BIOMARKER, or PATHWAY_MEMBER). This table summarizes up to 50 top-ranked drug candidates (25 sensitive and 25 resistance when available) derived from the mutational landscape of all samples included in the study. A detailed per-clone breakdown is available in the Drug Prioritization section.

Analysis mode: Clinical (standard filtering applied — experimental-status drugs and pathway-member interactions excluded).

The response of each drug is evaluated using the DScore (range -1 to +1):

- Sensitivity (DScore > 0): Positive values indicate that the mutations confer sensitivity to the drug (optimal candidates with a DScore > 0.7 are highlighted in bold).
- Resistance (DScore < 0): Negative values suggest that the clone possesses mutations associated with therapeutic resistance.

Drug-Gene Interactions:

- DT (Direct Target): The drug acts directly on the mutated protein.
- BM (Biomarker): The alteration acts as a biomarker predictive of response.
- PM (Pathway Member): The mutation affects the signaling pathway targeted by the drug.

The table below provides a simplified overview of the drugs targeting the affected clones and the specific genes. Drugs are prioritized using the following criteria: (1) clone coverage, (2) maximum dScore of the drug regarding the targeted genes (dScore), and (3) interaction type importance (DT>BM>PM).

The source files are available at: results/dream_P10/report/components/drug_summary.tsv.

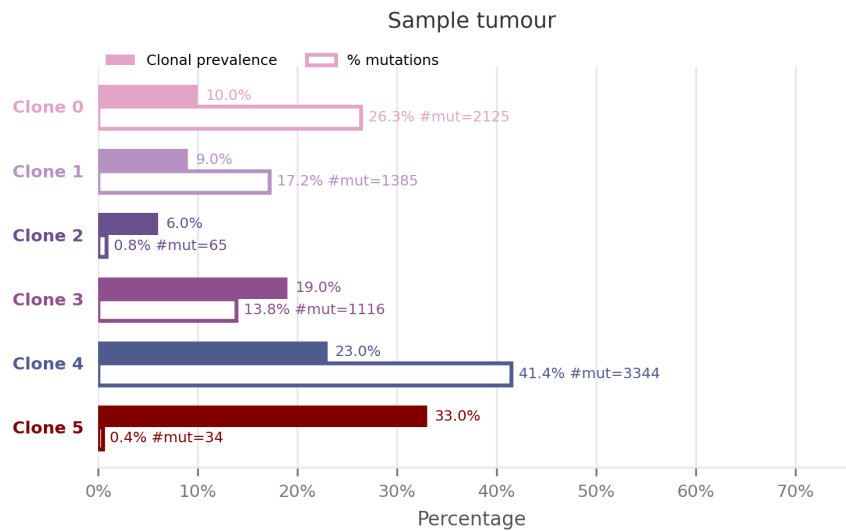
Drug	Status	Max. dScore	Target Clones	Gene Interactions	Drug Response
7440-50-8	APPROVED	0.712	Total: 3 DT: 0, 1, 4	DT: C1S, HEPHL1, P4HB	SENSITIVITY
DRONEDARONE	APPROVED	0.312	Total: 3 DT: 1; PM: 3, 4	DT: ADRA2C; PM: MED12L	SENSITIVITY
HYDROMORPHONE	APPROVED	0.715	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY

Drug	Status	Max. dScore	Target Clones	Gene Interactions	Drug Response
DIPHENOXYLATE	APPROVED	0.712	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
LOPERAMIDE	APPROVED	0.712	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
NALOXONE HYDROCHLORIDE	APPROVED	0.712	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
NALOXONE	APPROVED	0.316	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
NALTREXONE	APPROVED	0.316	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
CODEINE	APPROVED	0.315	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
FENTANYL	APPROVED	0.315	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
HYDROCODONE	APPROVED	0.315	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
MORPHINE	APPROVED	0.315	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
NALBUPHINE	APPROVED	0.315	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
BUTORPHANOL	APPROVED	0.314	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
NALMEFENE	APPROVED	0.314	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
OXYCODONE	APPROVED	0.314	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
ALVIMOPAN	APPROVED	0.313	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
BUPRENORPHINE	APPROVED	0.313	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
NALDEMEDINE	APPROVED	0.313	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
OXYMORPHONE	APPROVED	0.313	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
SAMIDORPHAN	APPROVED	0.313	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
SUFENTANIL	APPROVED	0.313	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
TRAMADOL	APPROVED	0.313	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY
7440-66-6	APPROVED	0.312	Total: 2 DT: 0, 1	DT: C1S, P4HB	SENSITIVITY
AMISULPRIDE	APPROVED	0.312	Total: 2 DT: 0, 1	DT: OPRD1	SENSITIVITY

Sample-Level Clonal Structure Analysis

Analysis of clonal evolution patterns and proportions across samples.

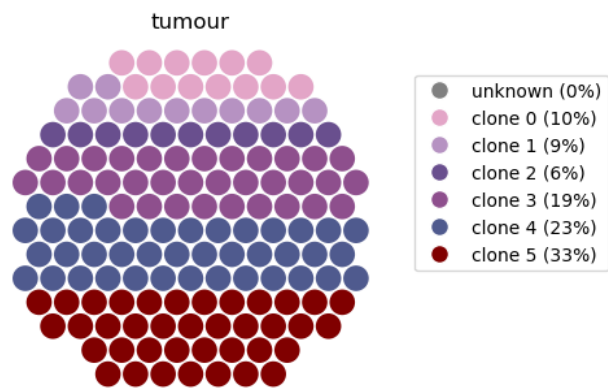
The source files are available at: results/dream_P10/report/components/mutation_contribution/{sample_id}_mutation_contribution.png.



Clonal Proportions

Based on the inferred tumor phylogeny, the spheres representations show the distribution and percentage of each clone across the different biopsies or time points, allowing us to visualize which clones dominate the tumor tissue.

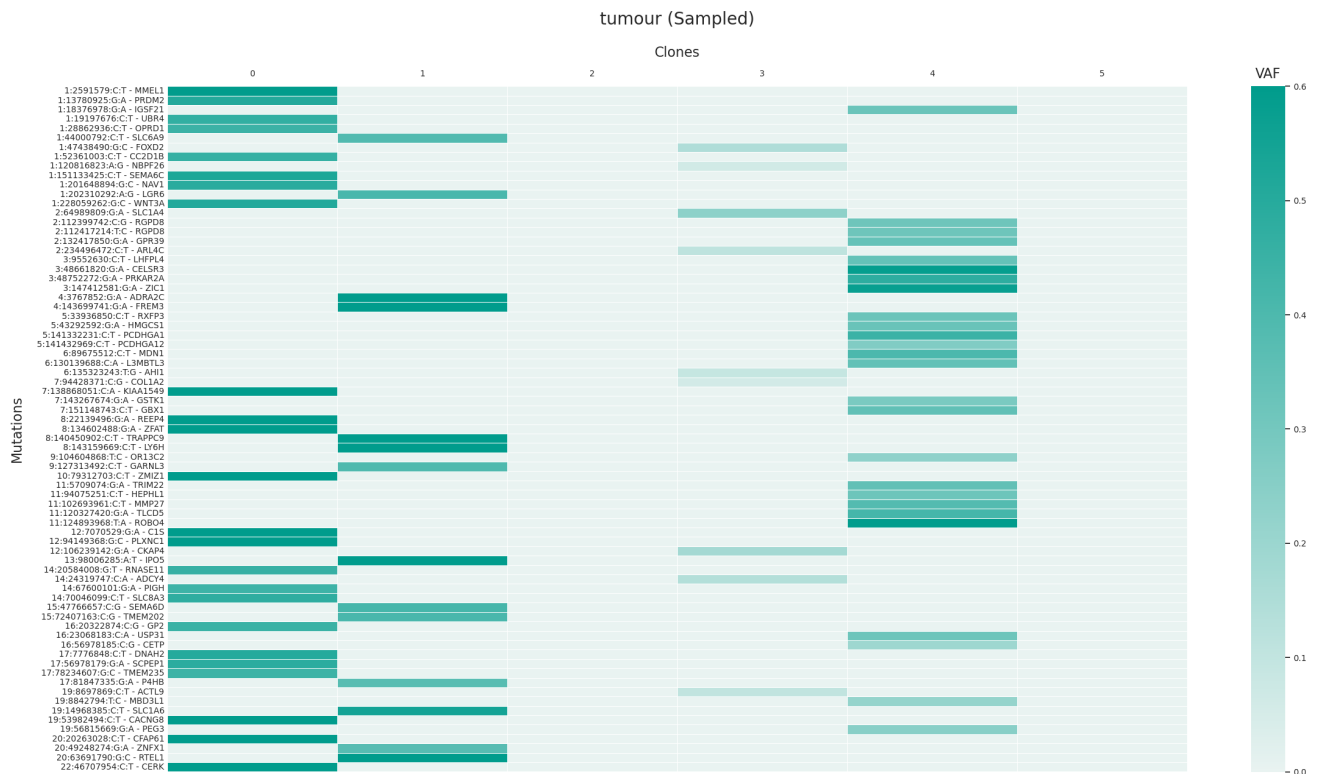
The source files are available at: results/dream_P10/report/components/spheres_of_clones/{sample_id}_sphere_of_clones.png.



Clonal Alterations

Heat maps illustrate the spatial or temporal evolution of mutations within clones. Observing how the frequency of a mutation varies across different samples allows us to distinguish early “trunk” events (present in all cells and responsible for the origin of the tumor) from later “branch” events. Identifying these branches is essential for understanding intratumoral heterogeneity and detecting emerging clones that may be driving disease progression or treatment resistance. Only variants with moderate or high predictive impact are shown.

The source files are available at: [results/dream_P10/report/components/vaf_heatmaps/sampled/{sample_id}_sampled_vaf_heatmap.png](https://results.dream_P10/report/components/vaf_heatmaps/sampled/{sample_id}_sampled_vaf_heatmap.png).



Pharmacogenomics

Gene Alterations

Both SNVs and small indels (if selected at the beginning of the workflow) used in clone inference are displayed with the most relevant information at the clone and sample level. To sum up the results, this section only shows top 10 mutations per clone with moderate or high impact.

Mutations are sorted using the following criteria:

- Clone.
- Mutation position.
- Impact: High and Moderate.

The source files are available at: [results/dream_P10/report/components/gene_alterations.tsv](#).

Clone	Mutation ID	Gene Symbol	Ensembl ID	VAF	Consequence	Impact
0	1:2591579:C:T	MMEL1	ENSG00000142606	tumour: 0.598	missense_variant	MODERATE
0	1:13780925:G:A	PRDM2	ENSG00000116731	tumour: 0.507	missense_variant	MODERATE
0	1:19197676:C:T	UBR4	ENSG00000127481	tumour: 0.470	missense_variant	MODERATE
0	1:28862936:C:T	OPRD1	ENSG00000116329	tumour: 0.444	missense_variant	MODERATE
0	1:52361003:C:T	CC2D1B	ENSG00000154222	tumour: 0.464	missense_variant	MODERATE
0	1:151133425:C:T	SEMA6C	ENSG00000143434	tumour: 0.525	missense_variant	MODERATE
0	1:201648894:G:C	NAV1	ENSG00000134369	tumour: 0.494	missense_variant	MODERATE
0	1:228059262:G:C	WNT3A	ENSG00000154342	tumour: 0.512	missense_variant	MODERATE
0	7:138868051:C:A	KIAA1549	ENSG00000122778	tumour: 0.613	missense_variant	MODERATE
0	8:22139496:G:A	REEP4	ENSG00000168476	tumour: 0.872	missense_variant	MODERATE
1	1:44000792:C:T	SLC6A9	ENSG00000196517	tumour: 0.389	missense_variant	MODERATE
1	1:202310292:A:G	LGR6	ENSG00000133067	tumour: 0.411	missense_variant	MODERATE
1	4:3767852:G:A	ADRA2C	ENSG00000184160	tumour: 0.755	missense_variant	MODERATE
1	4:143699741:G:A	FREM3	ENSG00000183090	tumour: 0.829	missense_variant	MODERATE
1	8:140450902:C:T	TRAPPC9	ENSG00000167632	tumour: 0.845	missense_variant	MODERATE
1	8:143159669:C:T	LY6H	ENSG00000176956	tumour: 0.729	missense_variant	MODERATE
1	9:127313492:C:T	GARNL3	ENSG00000136895	tumour: 0.397	missense_variant,NMD_transcript_variant	MODERATE
1	13:98006285:A:T	IPO5	ENSG00000065150	tumour: 0.820	missense_variant	MODERATE
1	15:47766657:C:G	SEMA6D	ENSG00000137872	tumour: 0.418	missense_variant	MODERATE
1	15:72407163:C:G	TMEM202	ENSG00000187806	tumour: 0.411	missense_variant	MODERATE
3	1:47438490:G:C	FOXD2	ENSG00000186564	tumour: 0.148	missense_variant	MODERATE
3	1:120816823:A:G	NBPF26	ENSG00000273136	tumour: 0.061	missense_variant	MODERATE
3	2:64989809:G:A	SLC1A4	ENSG00000115902	tumour: 0.235	missense_variant	MODERATE
3	2:234496472:C:T	ARL4C	ENSG00000188042	tumour: 0.111	missense_variant	MODERATE
3	6:135323243:T:G	AHI1	ENSG00000135541	tumour: 0.092	missense_variant	MODERATE
3	7:94428371:C:G	COL1A2	ENSG00000164692	tumour: 0.061	missense_variant	MODERATE
3	12:106239142:G:A	CKAP4	ENSG00000136026	tumour: 0.176	missense_variant	MODERATE
3	14:24319747:C:A	ADCY4	ENSG00000129467	tumour: 0.140	missense_variant	MODERATE
3	19:8697869:C:T	ACTL9	ENSG00000181786	tumour: 0.104	missense_variant	MODERATE
4	11:120327420:G:A	TLCD5	ENSG00000181264	tumour: 0.421	stop_gained	HIGH
4	11:124893968:T:A	ROBO4	ENSG00000154133	tumour: 0.600	stop_gained	HIGH
4	1:18376978:G:A	IGSF21	ENSG00000117154	tumour: 0.328	missense_variant	MODERATE
4	2:112399742:C:G	RGPD8	ENSG00000169629	tumour: 0.319	missense_variant	MODERATE
4	2:112417214:T:C	RGPD8	ENSG00000169629	tumour: 0.317	missense_variant	MODERATE
4	2:132417850:G:A	GPR39	ENSG00000183840	tumour: 0.338	missense_variant	MODERATE
4	3:9552630:C:T	LHFPL4	ENSG00000156959	tumour: 0.340	missense_variant	MODERATE
4	3:48661820:G:A	CELSR3	ENSG00000008300	tumour: 0.588	missense_variant	MODERATE
4	3:48752272:G:A	PRKAR2A	ENSG00000114302	tumour: 0.483	missense_variant	MODERATE
4	3:147412581:G:A	ZIC1	ENSG00000152977	tumour: 0.578	missense_variant	MODERATE

Drug Prioritization

The small variant analysis performed by PanDrugs2 displays only mutations classified as clinically relevant.

The vulnerability of each tumor clone to different drugs is assessed by considering only mutations with a relevant functional impact. Therapeutic options are prioritized by evaluating two fundamental dimensions: GScore (Biological Target) and DScore (Clinical Evidence). Check Drug Summary section to see how the scoring works.

We consider drugs with high clinical potential (DScore ≥ 0.7) that target key driver genes (GScore ≥ 0.6) to be Top Therapeutic Candidates (BTC), highlighted in bold.

The table below provides a simplified overview of the drugs targeting the affected genes, with the top 3 drugs selected per genetic alteration and ranked by Status and dScore. Drugs are sorted using the following criteria: (1) clone, (2) mutation position, (3) status, and (4) interaction type.

All definitions are provided in the Glossary of Terms.

The source files are available at: [results/dream_P10/report/components/gene_alterations.tsv](#).

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
0	12:7070529:G:A	C1S	tumour: 0.898	7440-50-8	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.345	CLINICAL_CANCER	DrugBank
				CONESTAT ALFA	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.312	0.345	nan	DrugBank
				7440-66-6	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.312	0.345	nan	DrugBank
0	1:28857272:T:C	OPRD1	tumour: 0.514	HYDROMORPHONE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.715	0.472	CLINICAL_CANCER	DrugBank; GuideToPharmacology; TEND; TTD; TdgClinicalTrial
				DIPHENOXYLATE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.472	CLINICAL_CANCER	DrugBank
				LOPERAMIDE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.472	CLINICAL_CANCER	DrugBank; TTD
0	1:28862936:C:T	OPRD1	tumour: 0.444	HYDROMORPHONE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.715	0.472	CLINICAL_CANCER	DrugBank; GuideToPharmacology; TEND; TTD; TdgClinicalTrial
				DIPHENOXYLATE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.472	CLINICAL_CANCER	DrugBank
				LOPERAMIDE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.472	CLINICAL_CANCER	DrugBank; TTD
0	1:74254707:C:A	TNNI3K	tumour: 0.446	FOSTAMATINIB	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.732	0.327	CLINICAL_CANCER	DrugBank
1	17:81847335:G:A	P4HB	tumour: 0.373	RALOXIFENE	APPROVED	BIOMARKER	UNCLASSIFIED	0.812	0.386	BREAST	DTC
				7440-50-8	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.386	CLINICAL_CANCER	DrugBank
				7440-66-6	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.312	0.386	nan	DrugBank
1	1:28844792:T:C	OPRD1	tumour: 0.365	HYDROMORPHONE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.715	0.472	CLINICAL_CANCER	DrugBank; GuideToPharmacology; TEND; TTD; TdgClinicalTrial
				DIPHENOXYLATE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.472	CLINICAL_CANCER	DrugBank
				LOPERAMIDE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.472	CLINICAL_CANCER	DrugBank; TTD
1	1:44000792:C:T	SLC6A9	tumour: 0.389	BITOPERTIN	CLINICAL_TRIALS	BIOMARKER	UNCLASSIFIED	0.013	0.325	nan	ChEMBLInteractions; GuideToPharmacology
1	1:74522939:G:A	TNNI3K	tumour: 0.377	FOSTAMATINIB	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.732	0.327	CLINICAL_CANCER	DrugBank
1	4:3767852:G:A	ADRA2C	tumour: 0.755	CABERGOLINE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.713	0.380	CLINICAL_CANCER	DrugBank; GuideToPharmacology
				CARVEDILOL	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.380	CLINICAL_CANCER	DrugBank
				CLOZAPINE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.380	CLINICAL_CANCER	DrugBank
3	2:64989809:G:A	SLC1A4	tumour: 0.235	ALANINE	CLINICAL_TRIALS	DIRECT_TARGET	UNCLASSIFIED	0.112	0.319	nan	DrugBank
3	5:33938127:C:T	RXFP3	tumour: 0.105	RELAXIN	CLINICAL_TRIALS	BIOMARKER	UNCLASSIFIED	0.012	0.196	nan	GuideToPharmacology
3	7:94428371:C:G	COL1A2	tumour: 0.061	OCRIPLASMIN	APPROVED	BIOMARKER	UNCLASSIFIED	0.212	0.244	nan	ChEMBLInteractions

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
4	11:102693961:C:T	MMP27	tumour: 0.383	MARIMASTAT	CLINICAL_TRIALS	DIRECT_TARGET	UNCLASSIFIED	0.512	0.196	CANCER	DrugBank
4	11:94075251:C:T	HEPBL1	tumour: 0.321	7440-50-8	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.319	CLINICAL_CANCER	DrugBank
4	16:56978185:C:G	CETP	tumour: 0.193	TAMOXIFEN	APPROVED	BIOMARKER	UNCLASSIFIED	0.812	0.325	BREAST	NCI
				ATORVASTATIN	APPROVED	BIOMARKER	UNCLASSIFIED	0.611	0.325	CLINICAL_CANCER	PharmGKB
				LOVASTATIN	APPROVED	BIOMARKER	UNCLASSIFIED	0.611	0.325	CLINICAL_CANCER	PharmGKB
4	5:33936850:C:T	RXFP3	tumour: 0.324	RELAXIN	CLINICAL_TRIALS	BIOMARKER	UNCLASSIFIED	0.012	0.196	nan	GuideToPharmacology
4	7:143267674:G:A	GSTK1	tumour: 0.284	GLUTATHIONE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.312	0.200	nan	DrugBank
				GLUTATHIONE SULFINATE	CLINICAL_TRIALS	DIRECT_TARGET	UNCLASSIFIED	0.112	0.200	nan	DrugBank

Glossary of Terms

Clone

A population of tumor cells that all share a specific set of mutations inherited from a common ancestor cell. Ideally, each inferred mutation cluster would correspond to a biologically distinct clone.

Cluster

Groups of cancer cells that share a common set of somatic mutations, which are inferred by clustering mutations with similar cellular prevalence estimates across one or more tumor samples.

Driver Gene

Genes whose mutations are responsible for the development and progression of cancer.

dScore

Pandrugs' indicator which measures the suitability of the treatment for a particular patient. It ranges from -1 to 1, with the negative values corresponding to resistance and the positive values corresponding to sensitivity.

gScore

Pandrugs' indicator which measures the biological relevance of a gene in the tumoral process and its druggability. It ranges from 0 to 1, with higher values corresponding to more relevant and actionable targets.

Interaction Type

Drugs approved for cancer treatment were manually classified into 7 different groups: chemotherapy, targeted therapy, hormone therapy, immunotherapy, photodynamic therapy, combination therapy, and other.

Pandrugs

A bioinformatics platform to prioritize anticancer drug treatments according to individual multi-omics data.

Parent cells

Group of cells that are set to be the origin of the tumor's phylogeny while performing the clustering and the clonal tree. The estimation of the proportions of parent cells may vary depending on the samples provided.

Study

Group of biologically related samples that are analyzed together in Clonucopy. The experimental design and grouping criteria are defined according to specific research objectives, allowing multiple use cases adapted to the needs of the comparative analysis.

Variant Allele Frequencies (VAF)

A biomarker that expresses the proportion of sequencing reads that support a specific variant allele relative to the total number of reads within a genomic locus. It ranges from 0 to 1.